

Single-Trajectory GRPO for Cooperative Multi-Agent Reinforcement Learning

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TL;DR — The n agents inside one episode form the GRPO group: critic-free returns globally over agents

1 Why critic-free MARL?

- PPO-style MARL (IPPO, MAPPO) trains a **critic + GAE**: extra memory, extra compute, and value-estimation error.
- **GRPO** drops the critic: advantages are rewards normalized across a *group of rollouts of the same prompt* — the recipe behind LLM post-training (DeepSeekMath).
- In MARL we do **not** get many rollouts from an identical state — so where does the group come from?
- **Our answer:** in homogeneous cooperative MARL, the n agents inside *one* episode already form the group.

2 Key idea: the agents are the group

Standard GRPO (LLM post-training)



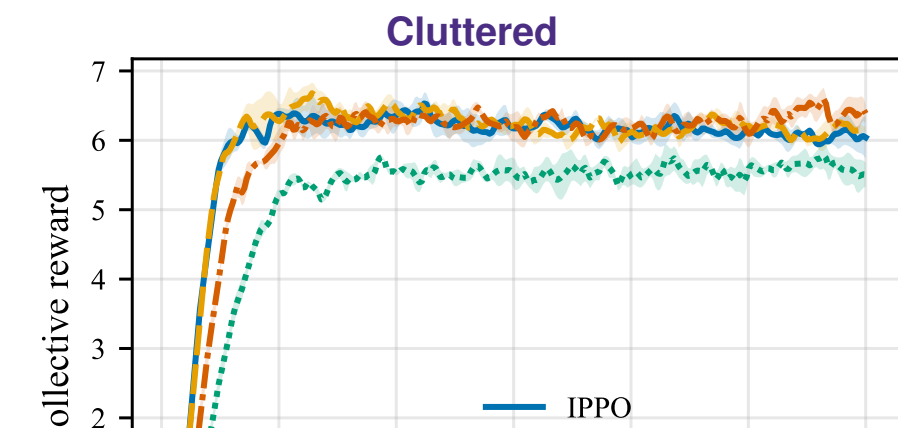
4 Setup: two sparse-reward games

Cluttered 15×15

30 fixed obstacles, fixed goal, randomized starts — moderate difficulty.

$n=3$ agents • egocentric $5 \times 5 \times 3$ obs • 7 actions • ≤ 100 reaching the goal: $r = 1 - \frac{\text{step_count}}{\text{max_steps}}$, else 0 • 3 seeds, mean

5 Main result — GRPO matches IPPO (RQ3)





Paper + code

github.com/ko120/IGRPO

Memory-free MARL matches PPO with 16% less peak GPU memory — normalize group statistics and time, not per time step.

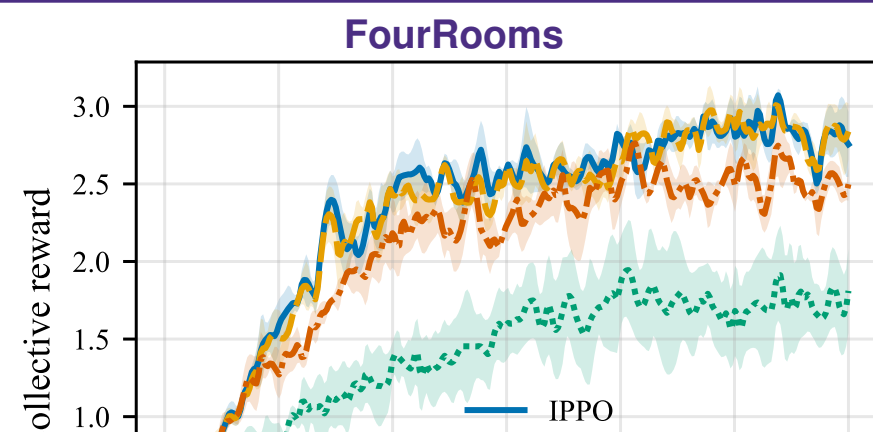
gridworlds

FourRooms 15×15

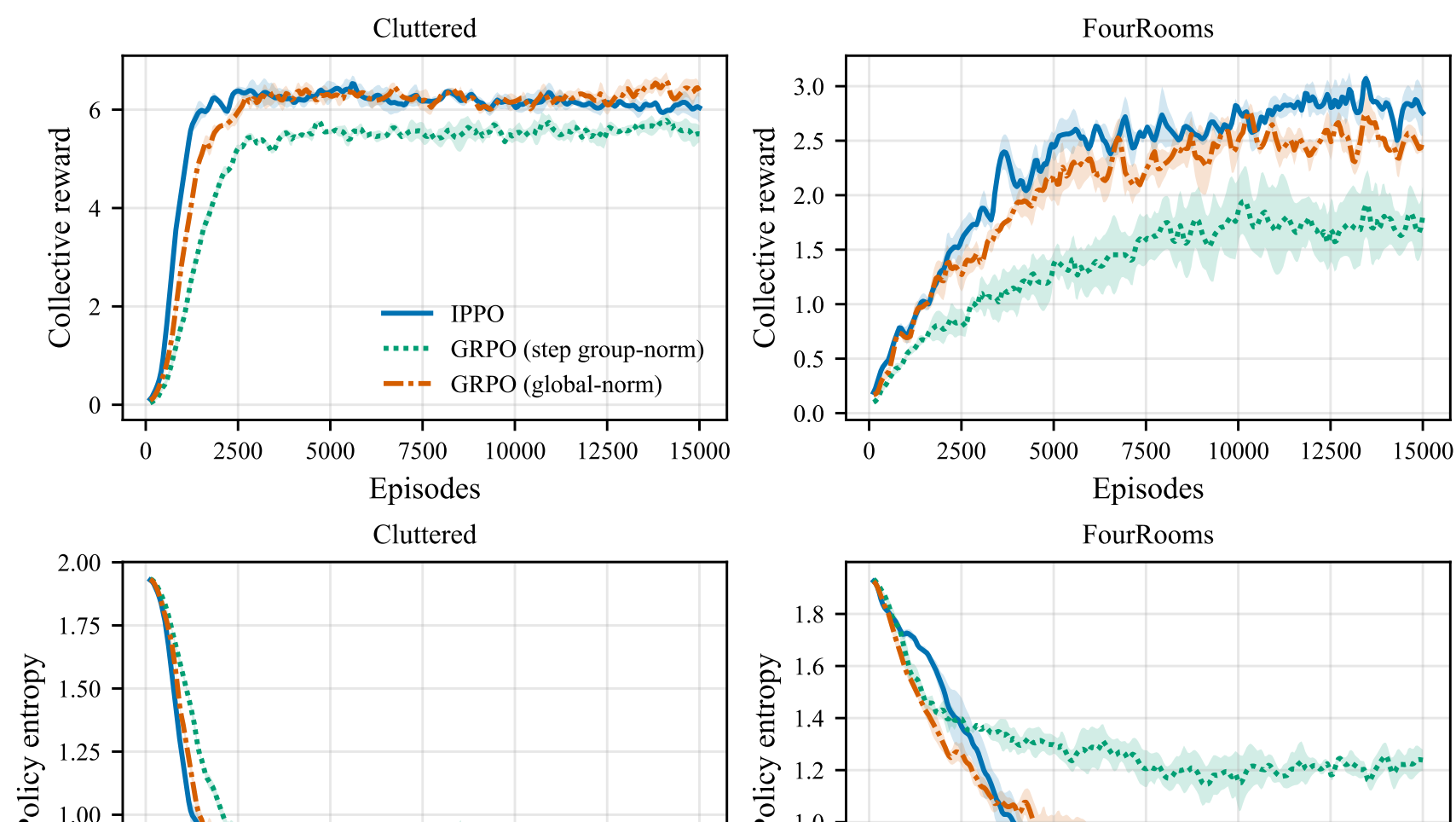
doorways, goal, *and* starts all randomized each reset — hard exploration.

100 steps • parameter sharing (CTDE) • reward only on goal, mean $\pm$ SEM • built on RLlib.

Matches PPO with less memory



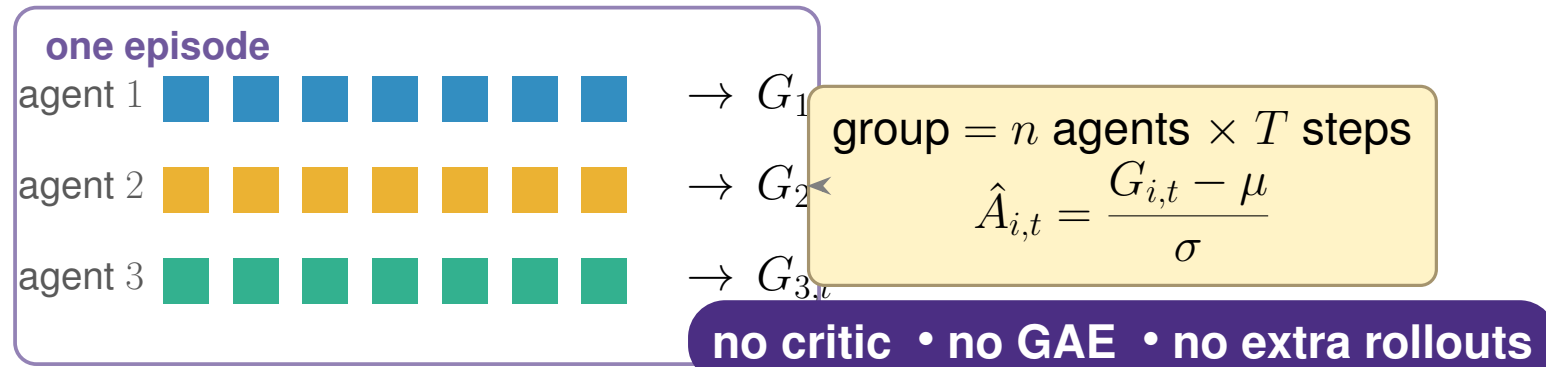
7 The group statistics are the critical choice (RQ4)





✗ needs many rollouts from the same state — unavailable in MARL

This work: single-trajectory GRPO



3 Method

PPO-like clipped objective, KL-regularized, critic-free:

$$J(\theta) = \mathbb{E}[\min(r \hat{A}, \text{clip}(r, 1-\epsilon, 1+\epsilon) \hat{A})] - \beta \text{KL}(\pi_\theta \parallel \pi_{\text{ref}})$$

value-loss coefficient 0, no entropy bonus, value targets $\equiv 0$. Advantages built from **discounted returns-to-go** $G_{i,t} = \sum_{\tau} \gamma^\tau R(s_{i,t+\tau}, u_{i,t+\tau})$.

A) Per-time-step cross-agent

(Zhao & Li, 2025)

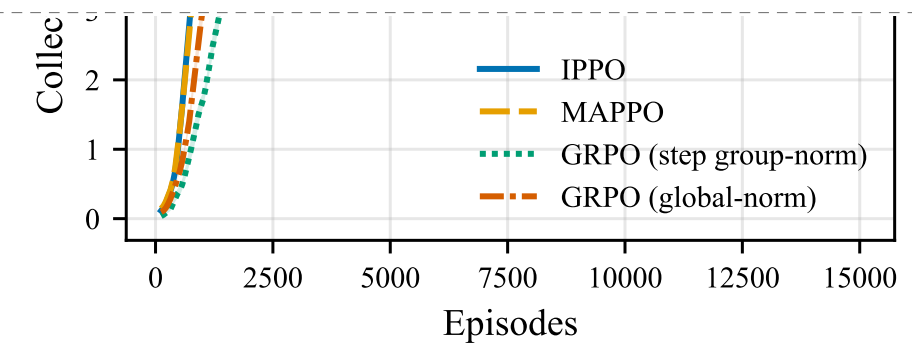
$\hat{A}_{i,t} = \frac{G_{i,t} - \mu_t}{\sigma_t}$, (μ_t, σ_t) over the n agents at step $t \Rightarrow$ only $n=3$ samples per step, and advantages are **zero-sum across agents at every step**.

B) Global episode statistics

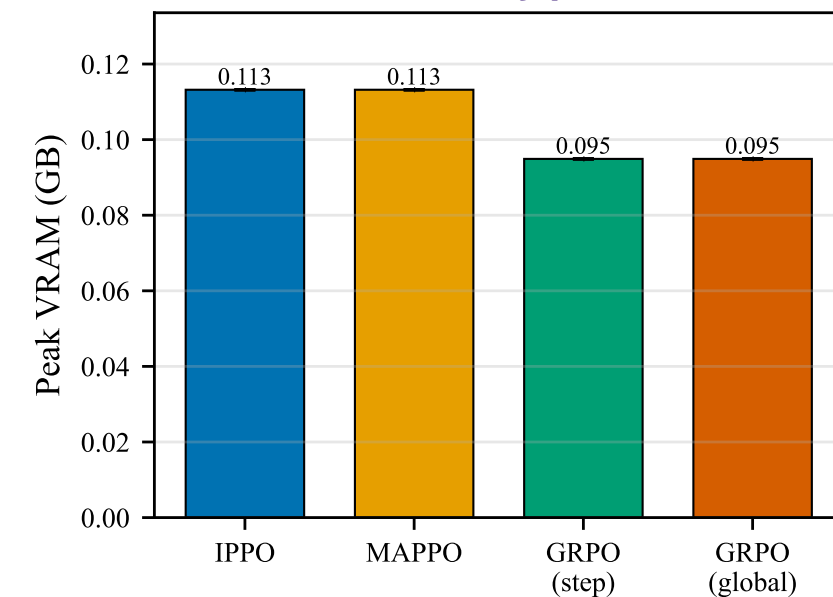
(this work)

$\hat{A}_{i,t} = \frac{G_{i,t} - \mu}{\sigma}$, one (μ, σ) over *all* nT values $\{G_{i,t}\} \Rightarrow nT$ -sample baseline that **preserves absolute return differences** within the episode.

Drop-in: swaps RLLib's GAE connector for a group-normalization connector — everything else is standard PPO machinery.



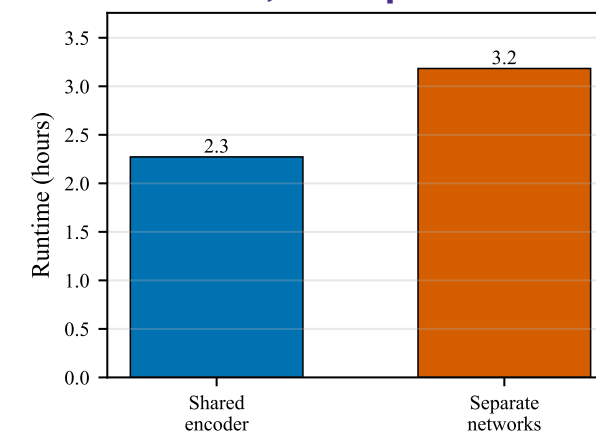
Peak GPU memory per run



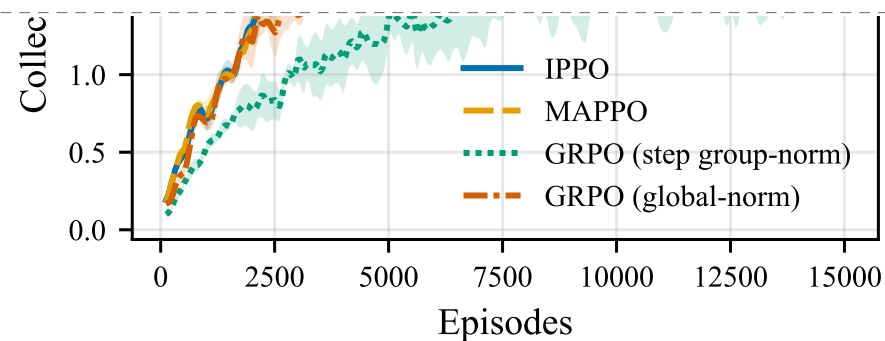
Takeaway: GRPO (global-norm) reaches IPPO critic — comparable wall-clock, 16% lower peak

6 PPO design choices (RQ1–RQ4)

Runtime, 30k episodes



- **Shared encoder** efficiency, but ϵ
- **MAPPO = IPP** critic adds no n both). More info



—16%
peak VRAM, critic-free

0.113 GB $\rightarrow$ 0.095 GB

6.4 ± 0.2 vs. 6.1 ± 0.1

GRPO (global) vs. IPPO — Cluttered

2.5 ± 0.1 vs. 2.8 ± 0.1

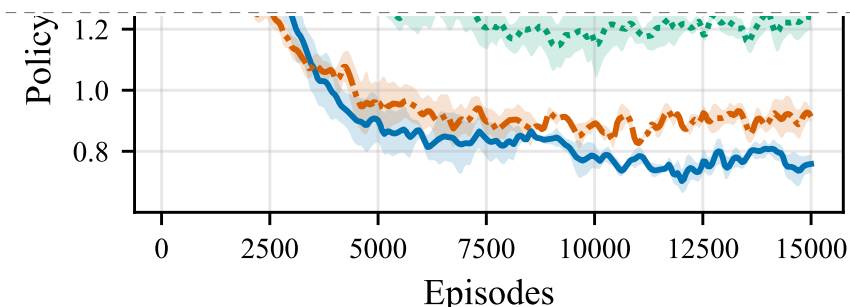
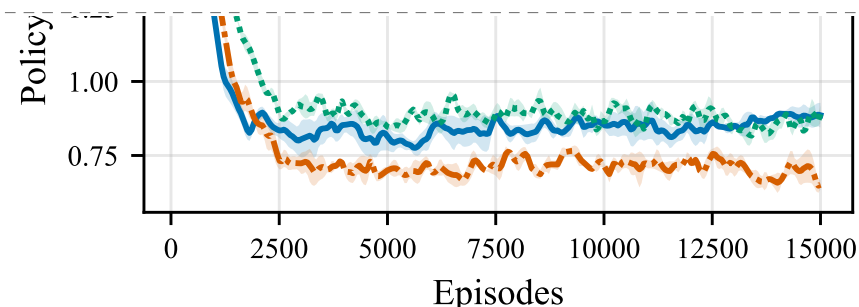
GRPO (global) vs. IPPO — FourRooms

IPPO-level reward on both environments with **no** peak VRAM at this model scale.

-RQ2)

encoder $\approx$ separate networks in sample but 30% **less wall-clock** (2.3 vs. 3.2 h).

IPPO under sparse rewards: the centralized has no measurable benefit (6.1 ± 0.1 / 2.8 ± 0.1 for π information $\neq$ better).



Top: collective reward. Bottom: policy entropy. Cluttered (left), FourRooms (right).

- On Cluttered both variants learn; on FourRooms **per-time-step collapses:** plateaus at 1.7 ± 0.3 (vs. IPPO 2.8 ± 0.1) with entropy stuck high — **global** stays competitive (2.5 ± 0.1).
- **Why:** (μ_t, σ_t) are estimated from only $n=3$ returns per step; under sparse reward the three returns are often identical (all zero) $\Rightarrow$ advantage collapses to zero — no learning signal.
- **Zero-sum grading:** even when *all* agents succeed, slower agents get negative advantage.
- Global pools nT samples into one stable baseline; with lateness-scaled rewards bounded in $[0, 1]$, return scales stay comparable across time, so one global mean is a sound baseline.

Takeaways

- ✓ **Critic-free GRPO is a practical CTDE recipe:** matches IPPO/MAPPO on both tasks with 16% less peak VRAM.
- ✓ **Normalize globally** over agents $\times$ timesteps — per-time-step cross-agent statistics fail on hard exploration.
- ✓ **Centralization is not free lunch:** MAPPO's global critic adds nothing under sparse rewards; shared encoders cut runtime 30%.

Limitations: two gridworlds from one family, $n=3$, small CNNs (absolute VRAM saving modest); formal variance analysis of single-trajectory baselines left to future work.